The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

3 High-Fidelity Electromagnetic Transient (EMT) Systems and IEEE 39-Bus Testing

To ground the continuous-time physics layer into a verifiable industrial framework, sub-millisecond electromagnetic transient (EMT) numerical simulations are executed across a modified IEEE 39-bus test system via the open-source Julia platform `PowerSimulationsDynamics.jl`. Let the transient state of an electrical node i be characterized by a continuous differential-algebraic power vector $\boldsymbol{\xi}_i(t)$:

$$\boldsymbol{\xi}_i(t) = [V_i(t),\, \delta_i(t),\, P_{\text{IBR},i}(t),\, Q_{\text{IBR},i}(t)] \in \mathbb{R}^4 \tag{2}$$

where $V_i(t)$ represents the instantaneous voltage magnitude, $\delta_i(t)$ defines the phase angle deviation, and $P_{\text{IBR},i}(t), Q_{\text{IBR},i}(t)$ monitor the active and reactive power outputs of local inverter-based renewable resources.

4 Horizontal Real-Time Control and Vertical MPC Optimization Plane

To isolate high-speed electromagnetic phase ruptures without optimization gradient collapse under severe phase noise, we establish an explicit operational split between the horizontal real-time execution lane and the vertical optimization plane. The horizontal path connects the error summing node directly to a dedicated sub-millisecond **Control** block. Symmetrically, the parameter estimator and the Model Predictive Control (**MPC**) module are positioned on the vertical plane.

The vertical optimizer tracks the joint physics-neural gray-box dynamics ($L_\theta \cdot \text{param}$) from partial measurements via a bi-level Moving Horizon Estimation (MHE) scheme following the paradigms of *Zhou & Köhler (2024)*. It projects a dynamically tightened state-space contraction multiplier $\alpha(\theta_{\text{NN}})$ down into the horizontal Control block:

$$\mathcal{X}(\theta_{\text{NN}}) = \alpha(\theta_{\text{NN}}) \cdot \mathcal{X}_{\text{nominal}} = \left(\frac{1}{1 + \beta \cdot \theta_{\text{NN}}} \right) \cdot \mathcal{X}_{\text{nominal}} \tag{3}$$

where $\beta \in \mathbb{R}^+$ defines the tuning gain and θ_{NN} tracks the unmapped phase-locked loop tracking errors. This in-series topology ensures the horizontal control lane restricts its actuation boundaries in real time, preventing the states from overdriving the unmodeled inner protective current rails.

4.1 Linear Matrix Inequality Formulation and Convergence Certification

Let $\boldsymbol{W}(x) = \boldsymbol{W}^T(x) > 0$ define the positive-definite dual contraction metric tensor. The system trajectories are certified stable under an L_2-gain noise attenuation index γ if the following symmetric matrix relation is strictly satisfied within the column boundaries:

$$\begin{bmatrix} \boldsymbol{\Theta}_{11}(x) & \boldsymbol{B}_w(x) \\ \boldsymbol{B}_w^T(x) & -\gamma^2 \boldsymbol{I} \end{bmatrix} \leq 0 \tag{4}$$

where the bounded sub-matrix component $\boldsymbol{\Theta}_{11}(x)$ is unrolled using a split block to prevent horizontal margin spills:

$$\begin{aligned} \boldsymbol{\Theta}_{11}(x) = &- \dot{\boldsymbol{W}}(x) + \boldsymbol{A}(x)\boldsymbol{W}(x) \\ &+ \boldsymbol{W}(x)\boldsymbol{A}^T(x) - 2\lambda \boldsymbol{W}(x) \end{aligned} \tag{5}$$

In this equation, $\boldsymbol{A}(x) = \frac{\partial f}{\partial x}$ acts as the system state Jacobian, $\boldsymbol{B}_w(x)$ maps the non-linear multiplicative phase noise vector parameters, and $\lambda \in \mathbb{R}^+$ defines the target exponential contraction convergence rate.

5 Black-Hunter-Pan Multi-Ontology Dialogue Strategy

When independent corporate data centers must resolve grid access boundaries without data leakage or brute-force database merges, the deconfliction process is modeled as a preference-ordered dialogue game following the algebraic principles of **Black, Hunter & Pan**. Let $\mathcal{D}_t$ represent the sequential state-history matrix of the active dialogue at turn t. The turn-taking strategy function $\mathcal{S}_x(\mathcal{D}_t)$ acts as a deterministic, preference-ordered selection transducer over the Description Logic poset, mapped via the following strict dominance hierarchy:

$$\mathcal{S}_x(\mathcal{D}_t) = \text{Scounter}_x(\mathcal{D}_t) \succ \text{Sarg}_x(\mathcal{D}_t) \succ \text{Sconcede}_x(\mathcal{D}_t) \succ \langle x, \text{close}, \gamma \rangle \tag{6}$$

Transmission bandwidth overhead $\mathcal{C}$ scales as a compact function of decision forest graph depth d rather than the raw magnitude of the extensional grounding data:

$$\mathcal{C}(\mathcal{T}) \propto \mathcal{O}(d \cdot \Sigma_{\text{tokens}}) \ll \mathcal{C}(\mathcal{E}) \propto \mathcal{O}(|\mathcal{E}| \cdot \text{Dim}_{\text{features}}) \tag{7}$$

This enables the multi-agent perimeter to achieve absolute strategic consensus $\langle x, \text{close}, \gamma \rangle \cap \langle y, \text{close}, \gamma \rangle$ over thin network channels while keeping proprietary Level 1 telemetry perfectly firewalled.